

Group-Fair Stochastic Gradient Descent via Masked Optimal Transport

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Table 1: stub

Lemma 1. *stub*

Lemma 2. *stub*

Theorem 3. *stub*

Theorem 4. *stub*

Corollary 5. *stub*

Theorem 6. *stub*

(1)

(2)

Proposition 7. *stub*

A APPENDIX

A. Proofs. *Proposition 7.* Conditionally on $A^t = A_j$ the within-group averages are unbiased for ∂f_i , so $\mathbb{E}[\text{batch subgradient} \mid A^t = A_j] = \sum_{i \in A_j} \mathbb{E}[\hat{\pi}_i^t \mid A_j] \partial f_i(\theta)$; averaging over q gives $\sum_i \tilde{p}_i \partial f_i = \partial F_{\tilde{p}}$, and $\sum_i \tilde{p}_i = \sum_j q_j \sum_{i \in A_j} \mathbb{E}[\hat{\pi}_i^t \mid A_j] = 1$. Both bounds follow from $0 \leq f_i \leq M$ and the two-point inequality $F_p(\theta_{\tilde{p}}^*) - F_p(\theta^*) \leq [F_p - F_{\tilde{p}}](\theta_{\tilde{p}}^*) + [F_{\tilde{p}} - F_p](\theta^*)$.

Sinkhorn scaling. Starting from any $T^{(0)} > 0$ on E with $\mathbf{1}^\top T^{(0)} = q$, alternate

$$T \leftarrow \text{diag}(p \oslash T \mathbf{1}) T, \quad T \leftarrow T \text{diag}(q \oslash T^\top \mathbf{1}) \quad (3)$$

until $\|T \mathbf{1} - p\|_1 \leq \varepsilon$. Under (1) the iteration converges to a solution of (??) [?, ?], being block-coordinate ascent on the dual of the entropic problem $\min\{\sum_E T \log T : T \mathbf{1} = p, \mathbf{1}^\top T = q\}$.

Lemma 1. Conditionally on $A^t = A_j$, $\mathbb{E}[g_i(\theta)] \in \partial f_i(\theta)$ for $i \in A_j$. Averaging over $A^t \sim q$, $\mathbb{E}[\hat{g}] = \sum_j q_j \sum_{i \in A_j} Y[i, j] \partial f_i(\theta) = \sum_i (\sum_j T[i, j]) \partial f_i(\theta) = \sum_i p_i \partial f_i(\theta)$, the row-marginal constraint being used in the last step.

Lemma 2. $Y[\cdot, j]$ is a probability vector on A_j , so $\|\sum_i Y[i, j] g_i\|^2 \leq \sum_i Y[i, j] \|g_i\|^2$ by convexity of $\|\cdot\|^2$; take expectations and use $\sum_j T[i, j] = p_i$: $\mathbb{E}\|\hat{g}\|^2 \leq \sum_i p_i \mathbb{E}\|g_i\|^2 \leq G^2$.

Theorem 6 (sketch). Route supply p_i from a source through group i , along the edges of E with infinite capacity, into composition j with demand q_j , and to a sink. A coupling exists iff the maximum flow saturates every source edge, which by max-flow/min-cut [?] is the right inequality of (1); the left one is the same statement for the reversed network. On singletons (1) is Hall’s condition [?].

Corollary 5. For $I = \{i\}$, $\mathcal{N}(I)$ is the set of compositions containing i , of total mass r_i , and $\mathcal{S}(I) = \{j : A_j = \{i\}\}$. Substitute $r_i = 1 - (1 - \pi_i)^b$ and $r_i = K/N$.

Theorem 3. By non-expansiveness of Π_C , $\|\theta^t - \theta^*\|^2 \leq \|\theta^{t-1} - \theta^*\|^2 - 2\eta \langle \hat{g}^t, \theta^{t-1} - \theta^* \rangle + \eta^2 \|\hat{g}^t\|^2$. Conditioning on θ^{t-1} , Lemma 1 and convexity bound the inner product below by $F_p(\theta^{t-1}) - F_p(\theta^*)$, and Lemma 2 bounds the last term by $\eta^2 G^2$. Summing, telescoping, and applying Jensen to $\bar{\theta}_T$ gives $\mathbb{E}F_p(\bar{\theta}_T) - F_p(\theta^*) \leq D^2/(2\eta T) + \eta G^2/2$, and the stated η balances the two terms [?]. In full: $\mathbb{E}\|\theta^t - \theta^*\|^2 \leq \mathbb{E}\|\theta^{t-1} - \theta^*\|^2 - 2\eta \mathbb{E}[F_p(\theta^{t-1}) - F_p(\theta^*)] + \eta^2 G^2$; summing over $t \leq T$, telescoping, dropping the terminal distance and applying Jensen to $\bar{\theta}_T$ gives $\mathbb{E}F_p(\bar{\theta}_T) - F_p(\theta^*) \leq D^2/(2\eta T) + \eta G^2/2$, which is $DG/\sqrt{T}$ at $\eta = D/(G\sqrt{T})$; the same bound holds for $\eta_t = \eta_0/\sqrt{t}$ up to constants.

Theorem 4. Add and subtract $F_p(\theta^\dagger) = \Phi(\tilde{p})$; the floor is exact by definition and $\Phi(p) = F_p(\theta^*)$. Theorem 3 applied to $F_{\tilde{p}}$ gives the residual bound on the surrogate; the residual on F_p differs from it by $(F_p - F_{\tilde{p}})(\bar{\theta}_T) - (F_p - F_{\tilde{p}})(\theta^\dagger)$, which vanishes as $\bar{\theta}_T \rightarrow \theta^\dagger$ (for instance under strong convexity, which both of our instances satisfy). The ℓ_1 bound is the two-point argument of Proposition 7 with w for $\tilde{p}$.

B. The infeasible regime in closed form (IMDb-Wiki, $\beta = 3$). On the mirrored instance, where 41 of 100 groups violate $p_i \leq r_i$ and hold 88% of the importance mass, the Sinkhorn limit truncates the target at the observation-rate ceiling, $\hat{p}_i \approx \min(p_i, r_i)$ up to renormalization (maximum

deviation 0.023), giving $\|p - \hat{p}\|_1 = 1.58$ of a maximum possible 2. Because the task is linear regression, the minimizers θ^* of F_p and $\theta^\dagger$ of $F_{\tilde{p}}$ are available in closed form: $\Phi(p) = F_p(\theta^*) = 82.9$ (matching the measured full-coverage plateau, 82.95) and $\Phi(\tilde{p}) = F_{\tilde{p}}(\theta^\dagger) = 105.9$. The alignment floor is therefore 23.0, and the realized bias equals its exact expression $|(p - \hat{p})^\top L(\theta^\dagger)| = 30.7$, where $L(\theta)$ collects the per-group risks. The ℓ_1 form $2M\|p - \hat{p}\|_1$ of Theorem 4 holds but is conservative here, since any uniform bound satisfies $M \geq \max_i L_i(\theta^\dagger) \approx 454$. The group-blind surrogate $\tilde{p}_i = r_i/K$ has the same floor to two decimals ($\Phi(\tilde{p}) = 105.9$); both surrogates starve the same groups, which is why Δ_{floor} vanishes at this endpoint of Table 1.

E. Instances and protocol, in full. Adult: logistic regression with $d=109$ features including an intercept, cross-entropy loss; 100 groups of 30 samples, each homogeneous in race, with 85/10/3/1/1 groups for White/Black/Asian-Pac./Amer-Ind./Other; the target gives each race 1/5, split equally among its groups, so on the prevalence instance the five groups of the three smallest races hold 60% of the target against $r_i \approx K/m = .03$. IMDb-Wiki: the first 100 identities with 30 images, 128-d ResNet embeddings, linear regression of age. Each step draws $K=3$ groups without replacement with probabilities $\propto r$, inducing q over the $\binom{100}{3}$ compositions; q is estimated from 10^6 draws and the plan fitted once per instance by (3), which on every instance with $\nu = 0$ reaches a row error below $3 \cdot 10^{-8}$, certifying the remaining inequalities of (1). All rules consume the same $H=5$ local updates from the same draw A^t . $\arg \min F_w$ is closed-form for least squares and Newton to machine precision for the logistic loss. Baselines: upsampling weights $\propto p_i/r_i$, downsampling $\propto \min(p_i/r_i, 1)$, LDS [?] the inverse Gaussian-smoothed age histogram (kernel 5, $\sigma=2$) averaged per group; each renormalized within the batch.

C. Adult severity sweep. The five Adult skews omitted from Table 1, same conventions.

β	ν	Δ_{floor}	gain	t
0	.60	.0110	+.0210	44.7
0.25	.60	.0144	+.0181	69.4
0.5	.40	.0129	+.0134	35.2
0.75	0	.0049	+.0050	35.8
1	0	.0003	+.0002	7.1

D. Adult per-race cross-entropy, prevalence regime. Tail-500 over 5 seeds, decaying stepsize.

	White	Black	Asian-Pac.	Amer-Ind.	Other
GAVOT	.339	.179	.368	.175	.073
group-blind avg.	.332	.200	.399	.189	.119
full	.379	.177	.328	.123	.031

Four of five categories improve and only the majority regresses, the expected effect of a category-uniform target; the residual on the two smallest is the unreachable mass of Cor. 5.